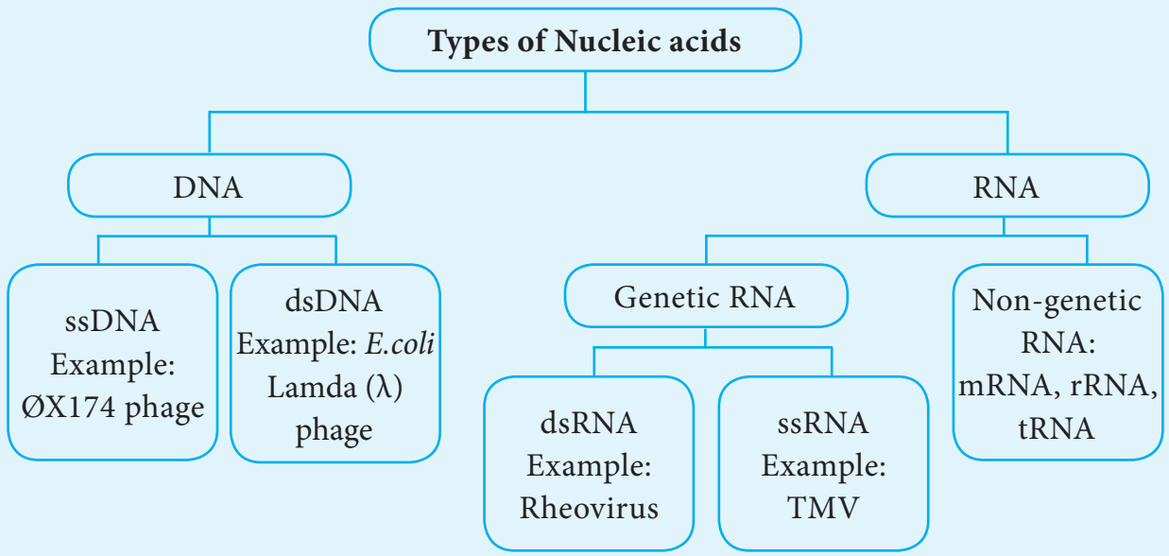



The entire genetic content of a cell is known as its genome and the study of genomes is genomics. In eukaryotic cells, but not in prokaryotes, DNA forms a complex with histone proteins to form chromatin, the substance of eukaryotic chromosomes. A chromosome may contain tens of thousands of genes. Many genes contain the information to make protein product. DNA controls all of the cellular activities by turning the genes "on" or "off."




which store genetic information in RNA. The understanding of DNA's role in heredity has led to variety of practical applications including forensic analysis, paternity testing and genetic screening.

## 14.2 DNA Structure

- DNA is a polymer of simple monomeric units, the nucleotides (Figure 14.6).
- Each nucleotide is made up of three components:
  1. Nitrogenous base
  2. Sugar
  3. Phosphate group
- Nucleotide without phosphate group is known as nucleoside.
- The sugar present in DNA is deoxyribose sugar.
- The nitrogenous bases present in DNA are
  - Purines – Adenine (A), Guanine (G)
  - Pyrimidines – Thymine(T), Cytosine (C)
- The nucleotide as a unit is formed by
  - Glycosidic bond between nitrogenous base and sugar,
  - Ester bond between phosphate group and sugar
- Each of the nucleotides is bonded by a phosphodiester bond to form a polynucleotide chain (strand) (Figure 14.7a).



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